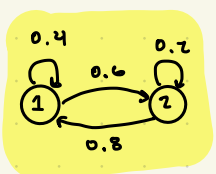
Assignment 7 Solutions

7.4

$$\begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ -2 & 3 & -13 & -1 & 8 \\ 2 & -4 & 18 & 3 & -2 \\ -3 & 5 & -22 & -4 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ 0 & 1 & -5 & -1 & -2 \\ 0 & -2 & 10 & 3 & 8 \\ 0 & 2 & -10 & -4 & -12 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ 0 & 1 & -5 & -1 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & -2 & -8 \end{bmatrix} \sim$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 2 & -2 & 1 & 0 \\ -3 & 2 & -2 & 1 \end{bmatrix} \qquad U = \begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ 0 & 1 & -5 & -1 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

8.1



A^1 has all positive entries

$$A - I = \begin{bmatrix} -0.6 & 0.8 \\ 0.6 & -0.8 \end{bmatrix} \sim \begin{bmatrix} -6 & 8 \\ 6 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & -4/3 \\ 0 & 0 \end{bmatrix}$$

$x_1 = \frac{4}{3} x_2$

x_2 is free

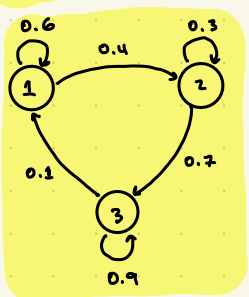
$x_1 + x_2 = 1 \Rightarrow$

$\frac{4}{3} x_2 + x_2 = 1 \Rightarrow$

$\frac{7}{3} x_2 = 1 \Rightarrow x_2 = \frac{3}{7}$

$\begin{bmatrix} 4/7 \\ 3/7 \end{bmatrix}$ is the unique steady state vector

8.2



A^2 has all positive entries

$$A - I = \begin{bmatrix} -0.4 & 0 & 0.1 \\ 0.4 & -0.7 & 0 \\ 0 & 0.7 & -0.1 \end{bmatrix} \sim \begin{bmatrix} -4 & 0 & 1 \\ 4 & -7 & 0 \\ 0 & 7 & -1 \end{bmatrix} \sim \begin{bmatrix} -4 & 0 & 1 \\ 0 & -7 & 1 \\ 0 & 7 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1/4 \\ 0 & 1 & -1/7 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = \frac{1}{4} x_3$

$x_2 = \frac{1}{7} x_3$

x_3 is free

$x_1 + x_2 + x_3 = 1 \Rightarrow$

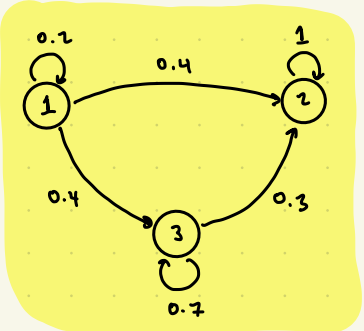
$\frac{1}{4} x_3 + \frac{1}{7} x_3 + x_3 = 1 \Rightarrow$

$\frac{39}{28} x_3 = 1 \Rightarrow$

$x_3 = \frac{28}{39}$

$\begin{bmatrix} 7/39 \\ 4/39 \\ 28/39 \end{bmatrix}$ is the unique steady state

8.5



A is not regular $A\vec{e}_2 = \vec{e}_2 \Rightarrow A^n\vec{e}_2 = \vec{e}_2$

$\Rightarrow A^n$ has zero entries

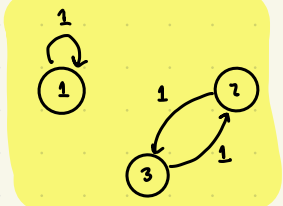
$$A - I = \begin{bmatrix} -0.8 & 0 & 0 \\ 0.4 & 0 & 0.3 \\ 0.4 & 0 & -0.3 \end{bmatrix} \sim \begin{bmatrix} -8 & 0 & 0 \\ 4 & 0 & 3 \\ 4 & 0 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ is the unique steady state vector

$x_1 = 0$, x_2 is free, $x_3 = 0$

$x_1 + x_2 + x_3 = 1 \Rightarrow x_2 = 1$

8.6



A is not regular

$A\vec{e}_1 = \vec{e}_1 \Rightarrow A^n\vec{e}_1 = \vec{e}_1 \Rightarrow A^n$ has zero entries

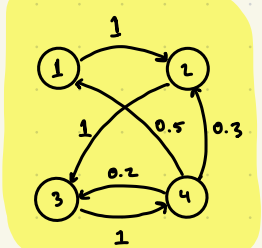
$$A - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

x_1 is free, $x_2 = x_3$, x_3 is free

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is a steady state vector

so is $\begin{bmatrix} 0.5 \\ 0.25 \\ 0.25 \end{bmatrix}$ (not unique)

8.13



$$A^6 = \begin{bmatrix} 1/10 & 3/20 & 23/100 & 3/50 \\ 3/50 & 19/100 & 39/125 & 159/500 \\ 23/50 & 3/25 & 109/500 & 81/125 \\ 3/10 & 23/50 & 3/25 & 109/500 \end{bmatrix}$$

A is regular

$x_1 + x_2 + x_3 + x_4 = 1 \Rightarrow$

$0.5x_4 + 0.8x_4 + x_4 + x_4 = 1$

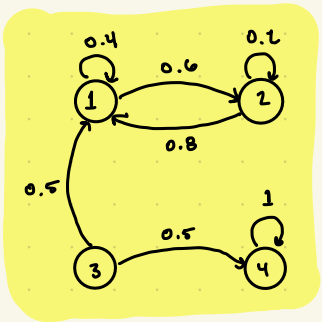
$3.3x_4 = 1 \Rightarrow x_4 = 10/33$

$$A - I = \begin{bmatrix} -1 & 0 & 0 & 0.5 \\ 1 & -1 & 0 & 0.3 \\ 0 & 1 & -1 & 0.2 \\ 0 & 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -0.5 \\ 0 & 1 & 0 & -0.8 \\ 0 & 1 & -1 & 0.2 \\ 0 & 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -0.5 \\ 0 & 1 & 0 & -0.8 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 0.5x_4$, $x_2 = 0.8x_4$, $x_3 = x_4$, x_4 is free

$\begin{bmatrix} 5/33 \\ 8/33 \\ 10/33 \\ 10/33 \end{bmatrix}$ is the unique SSV

8.14



$A\vec{e}_4 = \vec{e}_4 \Rightarrow A^k\vec{e}_4 = e_4 \Rightarrow A^k$ has zero entries

A is not regular

$$A - I = \begin{bmatrix} -0.6 & 0.8 & 0.5 & 0 \\ 0.6 & -0.8 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0.5 & 0 \end{bmatrix} \sim \begin{bmatrix} -6 & 8 & 5 & 0 \\ 6 & -8 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -4/3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 4/3\, x_2$
x_2 is free
$x_3 = 0$
x_4 is free

$x_1 + x_2 = 1$
$4/3\, x_2 + x_2 = 1$
$7/3\, x_2 = 1$
$x_2 = 3/7$

$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ is a S.S.V., so is $\begin{bmatrix} 4/7 \\ 3/7 \\ 0 \\ 0 \end{bmatrix}$ (not unique)

7.8 True. $A = LU$, L is invertible by def., $U = L^{-1}A \Rightarrow U$ is invertible and $U^{-1} = (L^{-1}A)^{-1} = A^{-1}(L^{-1})^{-1} = A^{-1}L$

8.17 False. $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $A - I \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ $x_1 = x_2$, x_2 is free, $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ is the unique SSV

$\lim_{k\to\infty} A^k \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ does not exist

8.19 True. $A^T\vec{1} = \vec{1} \Rightarrow (A - I)^T\vec{1} = (A^T - I)\vec{1} = \vec{1} - \vec{1} = \vec{0}$

$(A - I)^T$ has L.D. columns $\Rightarrow A - I$ has L.D. columns $\Rightarrow (A - I)\vec{x} = \vec{0}$ has nontrivial solution (by IMT)

That there's a nontrivial solution with nonnegative entries is difficult given what we know.

Challenge:

$A\vec{v} = \vec{v} \Rightarrow A|\vec{v}| = |\vec{v}|$ where $\left|\begin{bmatrix} v_1 \\ \vdots \\ v_k \end{bmatrix}\right| = \begin{bmatrix} |v_1| \\ \vdots \\ |v_k| \end{bmatrix}$

8.20 **True.** $\vec{1}^T \vec{v} = \vec{1}^T (A\vec{v})$

$$\vec{1}^T (AB)\vec{v} = \vec{1}^T \left(A(B\vec{v})\right) = \vec{1}^T (B\vec{v}) = \vec{1}^T \vec{v}$$

8.21 **False.** $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

8.22 **False.** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is not regular, but has unique S.S.V.

8.24 **False.** $T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ T is not regular, $\lim_{k\to\infty} T^k \vec{v} = \vec{v}$

8.28 (a) $T = \begin{bmatrix} p & (1-p) \\ (1-p) & p \end{bmatrix}$

(b) $T - I = \begin{bmatrix} (p-1) & (1-p) \\ (1-p) & (p-1) \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1-p}{p-1} \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ $x_1 = x_2$, x_2 is free

$$\lim_{k\to\infty} T^k \vec{e}_1 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

8.29 (a) $A = \begin{bmatrix} 1-p & p & 0 \\ 0 & 0.5-p & 0.7 \\ p & 0.5 & 0.3 \end{bmatrix}$ (b) $p = 0 \Rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.7 \\ 0 & 0.5 & 0.3 \end{bmatrix} \Rightarrow A^k \vec{e}_1 = \vec{e}_1$ not regular

(c) $A = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0 & 0.4 & 0.7 \\ 0.1 & 0.5 & 0.3 \end{bmatrix}$ $A - I \sim \begin{bmatrix} -1 & 1 & 0 \\ 0 & -6 & 7 \\ 1 & 5 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -7/6 \\ 0 & 6 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -7/6 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -7/6 \\ 0 & 1 & -7/6 \\ 0 & 0 & 0 \end{bmatrix}$

$x_1 = 7/6\, x_3$, $x_2 = 7/6\, x_3$, x_3 is free

$x_1 + x_2 + x_3 = 1$, $20/6\, x_3 = 1$, $x_3 = 6/20$

$$\lim_{k\to\infty} A^k \begin{bmatrix} 0.2 \\ 0.4 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 7/20 \\ 7/20 \\ 6/20 \end{bmatrix}$$

(d) $A - I = \begin{bmatrix} -0.5 & 0.7 \\ 0.5 & -0.7 \end{bmatrix} \sim \begin{bmatrix} 5 & -7 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -7/5 \\ 0 & 0 \end{bmatrix}$ $x_1 = \frac{7}{5} x_2$, x_2 is free $\begin{bmatrix} 7/12 \\ 5/12 \end{bmatrix}$ is a S.S.V. for the (2, 3) part of the diagram

$$\lim_{k \to \infty} A^k \begin{bmatrix} 0.2 \\ 0.4 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ (0.4)(7/12) \\ (0.4)(5/12) \end{bmatrix} = \begin{bmatrix} 12/60 \\ 28/60 \\ 20/60 \end{bmatrix}$$